Wavelet-Based Time-Frequency Analysis of Evolving fluids in a Microfluidic Rheometer

Subject:	Meeting with Supervisor
Group name/ identifier	Group 24
Meeting Location, Date & Time	Location: Not recorded Date: 13 August 2026 Time: Not recorded
Group members present	Rohan Mathew, Jianhao Zhang, Xuan Zhang, Ria Mishra, Xingcan Wang (Alex). Supervisor: Saumya.
Apologies	None

1 Actions from previous meetings

Task	**Who is responsible**	**Deadline**
Review client feedback and identify applicable techniques from the literature.	All members	Not recorded
Develop signal processing methods and investigate rheological data.	All members	Not recorded

2 Items discussed and decisions made

Item	**Decision (if applicable)**
Use of published models	Select relevant preprocessing and feature engineering techniques rather than adopting entire models. Compare methods using common error or cost measures.
Five phase analysis pipeline	Rohan demonstrated protocol classification, a 30 ms resampling grid, signal quality screening, Fourier/CWT/DWT analysis, synthetic benchmarking and real donor analysis. In the demonstration, 224 of 268 windows passed quality checks.
Features and controlled evaluation	The pipeline examined gel point estimates, spectral entropy and total harmonic distortion. A synthetic transition at 32 seconds provided a known reference because real blood recordings lacked exact clotting labels.
Method comparison and deployment	CWT ridgelines localized the synthetic transition effectively; DWT was fast but less suited to evolving signal shape. Five donor datasets and 24 real signals informed a preliminary deployment recommendation matrix.
Ablation and physics informed methods	Saumya advised adding or removing features incrementally to identify their value. Physics informed hybrid modelling, including differential equation residuals in a cost function, was discussed as a possible direction.
Team progress and schedule	Jianhao reviewed parameter tuning and step responses. Ria, Xuan and Xingcan were moving from literature review to implementation. Functional notebooks and quantitative results were requested for the next week.

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3 Agreed actions from this meeting

Task	**Who is responsible**	**Deadline**
Run small ablation studies to assess the contribution of individual features and wavelets.	All members	Next meeting
Apply relevant parameter tuning and step response techniques to the July 18 dataset.	Jianhao	Next week
Prepare working Python notebooks and quantitative analysis results.	All members	Next week
Explore how physical knowledge could be incorporated into model evaluation.	All members	Not recorded

4 Next meeting

Location	Not recorded
Date & Time	Next week; exact date and time not recorded
Goals	Review implementation results, ablation studies and progress on the July 18 dataset.

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